

Sampling-Based Predictive Control for Obstacle Avoidance with an Underactuated Triple-Pendulum Cartpole

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Abstract—We measure how far simple sampling-based model-predictive control gets on an obstacle-avoidance task with chaotic, underactuated dynamics: a three-link pendulum on a cart, one actuator for four degrees of freedom, that must drive through gaps narrower than the pendulum is long. Six MJPC planners are compared at equal iteration budget on a single bottleneck and on three chained bottlenecks, under a collision test that allows no penetration at all. On one bottleneck the best planner reaches 76/100; on three, the best reaches 4/100 and five of the six never succeed. A memoryless control that discards its incumbent each iteration solves none of 300 trials, placing the warm start rather than the update rule at the centre of what these planners do. We then sweep the two parameters of the avoidance cost — its weight and the clearance margin at which it activates — over 5×5 configurations, three seeds each, 50 trials per cell. The two do not act independently: the surface is a ridge, and following it raises the same planner from 2.7% to 54% on the three-bottleneck course. Extending the sweep by a further $64 \times$ in weight and down to a 0.01 m margin follows the ridge to 65% (95% CI 57–72), with the best margin sliding down as the weight rises and the gain per factor of four falling below what 150 trials per cell can resolve. Widening the search from 10 rollouts per iteration to 40 — one integer in the task file, and the budget annealed sampling already spends — adds a further 20 points to 85% (79–90), at $2.65 \times$ the wall time rather than $4 \times$ on a 20-core x86 host, because 10 rollouts could not fill the thread pool; on a 15-core Apple Silicon host the same change costs $5.1 \times$ and leaves the budget. Eighty adds nothing. At that matched budget the plain sampler beats annealed sampling by 22 points, and annealed sampling on 40 rollouts is indistinguishable from the plain sampler on 10, so the annealing schedule converts a fourfold budget into nothing measurable. Carrying the plan forward by shifting and extending it instead of resampling it is a wash. We report per-iteration cost, control effort and cost share for every configuration, and single-thread and multi-thread planning throughput on two hosts at both rollout counts. We also report a negative result for AgileRRT, the sample-based planner this task was originally built for, and for iLQG, whose finite-difference derivatives dominate its cost while its linearization is invalidated by contact. All figures in this report were produced by seeded runs whose exact command lines are recorded alongside them.

Index Terms—model predictive control, sampling-based optimization, obstacle avoidance, underactuated systems, MuJoCo

I. INTRODUCTION

Predictive control with a physics engine in the loop is now cheap enough to run online, and MJPC [1] makes the comparison between planners concrete: one task specification, one rollout budget, several optimizers behind the same interface. What it does not settle is which problems the derivative-free planners can actually close, and what determines the boundary.

This report characterizes that boundary on one task. The system is the three-link pendulum cartpole of Caldwell and Correll [3], which has four degrees of freedom, one actuator, chaotic passive dynamics, and an obstacle gap half the pendulum’s length. Passing the gap requires laying the pendulum out near-horizontal, driving through, and recovering — a maneuver that cannot be held open-loop and that a single contact disqualifies.

We report three things.

First, a planner comparison at fixed iteration budget on one bottleneck and on three, under a zero-tolerance collision test. The ordering is stable across both worlds, but the success rates are not: three chained gaps take the best planner from 76/100 to 4/100.

Second, a two-parameter sweep of the avoidance cost. The weight and the clearance margin are the only knobs in the cost that concern the obstacles, and they are not interchangeable: weight sets what a violation costs, margin sets how early the cost arrives. Mapped over 5×5 values with three seeds per cell, the success rate on the three-bottleneck course is not a monotone function of either: the surface is a ridge, and the best margin slides down as the weight rises. Following it past the first grid reaches 65%, an order of magnitude above the default.

Third, the width of the search itself. Holding the objective at the best ridge cell and raising the rollouts per planning iteration from 10 to 40 reaches 85%. This is a single integer in the task file rather than an algorithm, it is the second largest effect we measured, and it is the same rollout budget that annealed sampling has been spending on its annealing schedule throughout. Eighty rollouts adds nothing.

Fourth, the cost of running any of this: per-iteration wall time by planner, where that time goes, control effort and per-

term cost share for the surviving configurations, and throughput on an Intel Core Ultra 7 265KF and an Apple M5 Pro.

We also record two negative results, because both were measured rather than assumed: AgileRRT does not fit a receding-horizon budget by three orders of magnitude, and iLQG is the weakest planner here for reasons that are structural.

II. TASK

A. System

A cart of mass 1.0 kg slides on a horizontal rail. Attached to it is a three-link pendulum; each link is a massless rod of length $L = 1/3$ m carrying a point mass of 0.1 kg at its tip, modelled as a sphere of radius $\rho = 0.028$ m. The single actuator applies a horizontal force to the cart, $|u| \leq 20$ N. There is no joint damping and no friction. With $q = (x_{\text{cart}}, \theta_1, \theta_2, \theta_3)$ the system has $n_q = 4$, $n_v = 4$, $n_u = 1$. Angles follow the reference implementation: $\theta_i = 0$ points link i along $+z$, so head i sits at

$$p_i = \left(x_{\text{cart}} + L \sum_{j \leq i} \sin \theta_j, L \sum_{j \leq i} \cos \theta_j \right). \quad (1)$$

Physics is integrated by MuJoCo [2] with `implicitfast` at $\Delta t = 5$ ms.

B. Two worlds

Obstacles are cylinders of radius $R = 0.6$ m with axis along y , so in the xz plane they are disks. They are real collision geoms, not merely cost terms.

Corridor is the task of [3], §5: one pair of disks at $x = 3$, $z = \pm 0.85$, leaving a gap of $2(0.85 - 0.6) = 0.5$ m, and a goal at $x = 6$. The pendulum is 1.0 m long, so it cannot pass upright.

Slalom keeps that bottleneck unchanged and adds two more at $x = 6$ and $x = 9$, with the goal at $x = 11$. The gaps are identical and the spacing is 3 m, which is 0.8–1.0 s of travel at the speeds the cart reaches and therefore straddles the 1 s planning horizon. Because the first bottleneck is the corridor task’s exactly, the difference between results on the two worlds is attributable to what follows it.

C. Success criterion

A trial is a *solve* if the cart ends within 0.3 m of the goal and no head ever overlapped a disk. The pendulum’s terminal configuration is not scored, because the objective used throughout gives the upright term zero weight: requiring an upright finish would score a run against a term it was never given.

The collision test admits no penetration. An earlier version of this benchmark allowed 20 mm, on the theory that soft contacts permit a few harmless millimetres on touch. Two measurements retract it. Across 30 recorded rollouts, $n_{\text{con}} > 0$ and $\min_i d_i < 0$ agree on every step — MuJoCo never reports contact between separated geoms, so there is no grazing band to protect. And measured penetrations run continuously from 2 to 39 mm, so no depth threshold separates numerical from real. At 20 mm on a 28 mm head, a sphere

TABLE I
RESIDUAL TERMS. N IS THE NUMBER OF DISKS: 2 ON THE CORRIDOR, 6 ON THE SLALOM.

Term	dim	residual
Cart	1	$x_{\text{cart}} - x_{\text{goal}}$
Upright	3	$\cos \theta_i - 1$
Velocity	4	$\dot{q}$
Control	1	u
Avoidance	$3N$	$\max(0, \delta - d_{ij})$

71% inside a disk scored as clean. We report both criteria in Section V-B and V-C, because the gap between them is itself informative, but the zero-tolerance column is the result.

III. PREDICTIVE CONTROL FORMULATION

A. Objective

We follow MJPC’s formulation [1]. A task is specified by a residual vector $r(x, u)$ partitioned into M terms, each with a norm n_i and a weight w_i ; the running cost is

$$c(x, u) = \sum_{i=1}^M w_i n_i(r_i(x, u)), \quad (2)$$

and every term here uses the quadratic norm $n_i(r) = \frac{1}{2} \|r\|_2^2$. At each control step the planner minimizes the mean running cost over a horizon of T steps,

$$J(u_{0:T-1}) = \frac{1}{T} \sum_{t=0}^{T-1} c(x_t, u_t), \quad x_{t+1} = f(x_t, u_t), \quad (3)$$

with x_0 the current state and f one call to `mj_step`. The control sequence is not free: it is the sampling of a linear spline with $K = 12$ knots over the horizon, so the decision variable is $\theta \in \mathbb{R}^{Kn_u}$ and $u_t = \text{spline}(\theta; t)$. With $T_h = 1.0$ s this is 83 ms of knot spacing. The first control of the optimized sequence is applied, the spline is shifted forward, and the next iteration warm-starts from it.

B. Cost terms

Table I lists the five residual terms. The weights are stated on the command line for every result in this report, so each table says what was optimized.

The upright residual is written $\cos \theta_i - 1$ rather than θ_i because it is zero exactly at the upright equilibrium and smooth across the $\pm\pi$ wrap, so the planner never sees a discontinuity manufactured by angle unwrapping.

C. Avoidance residual

Clearance between head i and disk j is surface-to-surface in the xz plane,

$$d_{ij}(x) = \|p_i(x) - o_j\|_2 - R_j - \rho_i, \quad (4)$$

so $d_{ij} = 0$ means touching. The avoidance residual is a hinge on it,

$$r_{ij}^{\text{avoid}}(x) = \max(0, \delta - d_{ij}(x)), \quad (5)$$

with δ the *clearance margin*. Combined with the quadratic norm and weight w_{avoid} , the avoidance term contributes

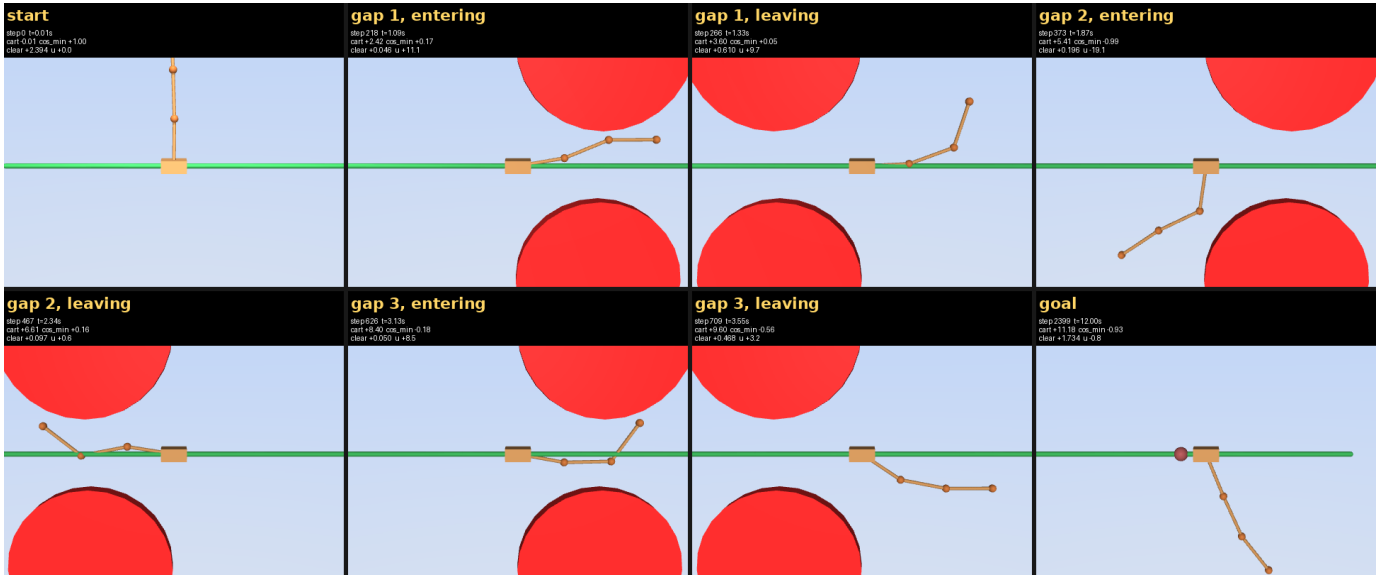


Fig. 1. One clean slalom solve, predictive sampling at avoidance weight 128,000 and margin 0.04 m, seed 1000. The camera tracks the cart, and panels are placed at the events that decide the run rather than at uniform intervals. Each panel reports the step, simulated time, cart position, $\min_i \cos \theta_i$, the minimum head-to-disk clearance $\min_i d_{ij}$ in metres, and the applied force. Top row: the start, upright; the swing-down at $t = 0.33$ s; the tightest moment of the run at $t = 1.12$ s, 27 mm of clearance inside the first gap; clear of it at $t = 1.20$ s. Bottom row: the second and third crossings at $t = 2.14$ and 3.31 s, both with 27 cm of clearance; the goal reached at $t = 4.82$ s; and the run parked at $t = 12$ s. At all three crossings the links are laid out along the rail at hub height, which is the posture that fits a 0.5 m gap — the cart passes the gap while the pendulum is horizontal, and the run is decided by whether the swing phase happens to place it there.

$\frac{1}{2} w_{\text{avoid}} \sum_{ij} \max(0, \delta - d_{ij})^2$: exactly zero when every head is more than δ from every disk surface, and growing quadratically inside that band.

Two properties of (5) matter for the rest of this report.

The term exists to give the sampler a gradient *before* contact. Contact alone is a discontinuity in the rollout cost, and a sampler that only learns about a disk by hitting it learns too late to steer around it. The cost is what converts the obstacle into something the sampler can see at a distance.

And δ and w_{avoid} do different things. The weight sets how much a violation costs; the margin sets how early the cost arrives, which is the only one of the two that buys the planner time. The half-gap is 0.25 m, so a margin approaching that makes the hinge active across the entire gap: even a head travelling exactly down the centreline is inside the penalty, and the barrier around each disk becomes a wall across the corridor. Section V-D measures where that transition happens.

D. Planners: one loop, three branch points

All six planners share the interface, the rollout budget, and the objective. Five of the six also share a loop, and it is worth writing that loop out because the differences between them are smaller than their names suggest.

Start from the simplest. *Random Sampling* has no state: every iteration it draws n control splines from a fixed Gaussian centred on zero control, rolls each out, keeps the cheapest, applies its first control, and throws the rest away. There is no warm start and nothing carries between iterations. It is a control rather than a proposal — the gap between it and any planner that does keep state is the measured value of keeping

Algorithm 1 One planning iteration, shared by five of the six planners.

Input: incumbent θ , state x_0 , budget n , horizon T

- 1: $\bar{\theta} \leftarrow \text{CENTRE}(\theta)$ $\triangleright$ what to sample around
- 2: **for** $k = 1 \dots n$ **do**
- 3: $\epsilon^{(k)} \sim \mathcal{N}(0, \Sigma^{(k)})$ $\triangleright$ SPREAD
- 4: $\theta^{(k)} \leftarrow \bar{\theta} + \epsilon^{(k)}$
- 5: $u_t \leftarrow \text{spline}(\theta^{(k)}; t)$, roll out f from x_0
- 6: $J^{(k)} \leftarrow \frac{1}{T} \sum_{t < T} c(x_t, u_t)$ $\triangleright$ Eq. (3)
- 7: **end for**
- 8: $\theta \leftarrow \text{FOLD}(\{\theta^{(k)}\}, \{J^{(k)}\})$ $\triangleright$ how the winner is kept
- 9: apply $\text{spline}(\theta; 0)$; shift θ forward one step

state.

Predictive Sampling [1] changes exactly one thing: it centres the draw on the incumbent θ rather than on zero, and keeps the incumbent itself as one of the n candidates so the update can never move uphill. Every remaining sampling planner here is a further change to one of three choices in the same loop — where the samples are centred, how they are spread, and how the winner is folded back into the incumbent. Algorithm 1 writes the loop with those three as named functions; Table II gives each planner’s instantiation.

Cross-Entropy [4] changes CENTRE, SPREAD and FOLD together: it takes the elite fraction of the sample set, fits a Gaussian to it, and makes that fitted Gaussian both the next centre and the next spread. It is the only planner here whose covariance is estimated from data rather than set.

Particle Swarm Optimization [5] changes CENTRE: each

TABLE II
HOW EACH PLANNER INSTANTIATES ALGORITHM 1. σ IS THE SAMPLING
STANDARD DEVIATION IN NEWTONS AGAINST A ± 20 N ACTUATOR.
ROLLOUTS ARE PER PLANNING ITERATION.

planner	CENTRE	SPREAD	FOLD	roll.
Random Sampling	0	$\sigma = 8$	$\arg \min_k$	10
Predictive Sampling	θ	$\sigma = 8$	$\arg \min_k$	10
Cross-Entropy	elite mean	elite cov.	elite mean	10
PSO (fixed)	$\theta + v$	$\sigma = 4$	$\arg \min_k$	10
PSO (stock)	$\theta + v$	$\sigma = 4$	post-update	10
Annealed Sampling	$\theta, 4\times$	annealed	MPPI mean	40

candidate carries a velocity v that drifts it toward its own best position and the swarm’s. Two variants are reported because they differ in FOLD. The stock variant publishes the post-swarm-update iterate, whose cost was never evaluated; the fixed variant publishes the best evaluated candidate. One implementation detail matters more than either: the particle state is re-initialized at the top of every planning iteration, so positions and velocities never persist and what remains is a single Gaussian round with a hard-coded $\sigma = 0.1 \times \text{ctrlrange} = 4.0$ N — half what Predictive Sampling uses. That halved spread, not the swarm, is what Section V-B measures.

Annealed Sampling implements DIAL-MPC’s dual-loop schedule [6]: it runs the whole of Algorithm 1 $N = 4$ times per planning iteration, re-centring on the updated incumbent each time, with an MPPI-style weighted mean [7] as FOLD instead of $\arg \min$. Its SPREAD is scheduled in two directions at once,

$$\Sigma_h^i = \exp\left(-\frac{N-i}{\beta_1 N} - \frac{H-h}{\beta_2 H}\right)I, \quad (6)$$

narrowing over the N stages (i) and wider for far-horizon knots than near-horizon ones (h). It therefore costs $4\times$ the rollouts of every other planner — 40 against 10 — and no setting of the harness equalizes that, because the harness holds *iterations* constant, not rollouts.

iLQG [8] is not in this family. It is the one first-order method: it builds finite-difference model and cost derivatives around a nominal trajectory, solves the Riccati recursion backwards for a time-varying affine policy, and line-searches forward. It has no SPREAD to set and draws no samples, which is why it appears in none of the columns above and why Section V-J reports its per-iteration cost so differently.

E. AgileRRT

The task comes from [3], whose planner is a kinodynamic RRT [9] with a steering primitive obtained by linearizing about the zero-control trajectory $x_{\text{zero}}(t)$ from a vertex rather than about the vertex point x_0 . Steering to a target is a controllability-Gramian problem on that linearization, projected back onto the true dynamics. We implemented the primitive against MuJoCo and grew trees with it, both to have the paper’s own method as a baseline and to test whether it

could serve as an MJPC planner. Section V-A reports what it does.

IV. PROTOCOL

A. Budget and the two clocks

Control is applied once per timestep of *simulated* time — 200 Hz — at every setting. The harness’s `--speed` flag changes the wall clock: at `speed = s` the loop runs at $200s$ Hz in wall time and the planner gets $1/s$ iterations per control step. Every result below uses $s = 0.25$, so the system is controlled at 50 Hz with four optimization iterations per decision. The invariant worth keeping is that the wall-clock budget per iteration is one timestep, 5 ms, at any speed; it is not the control period, which at $s = 0.25$ is 20 ms.

Holding *iterations* rather than seconds constant is what makes the comparison one of algorithms rather than of single-thread throughput. What an iteration costs each planner is reported separately, in Section V-J.

B. Trials, seeds, and objective

Each trial perturbs the initial state with Gaussian noise of scale 0.02 from a fixed stream (`--seed=1`), so trial k is the same start for every planner. Without this a deterministic planner such as iLQG would report the same run 100 times and its success rate would be 0 or 1 by construction.

The planner comparisons of Sections V-B and V-C use 100 trials, 20 s episodes, and the objective

$$w = (w_{\text{cart}}, w_{\text{up}}, w_{\text{vel}}, w_{\text{ctrl}}, w_{\text{avoid}}) = (1, 0, 0.1, 0.01, 500) \quad (7)$$

with margin $\delta = 0.08$ m, the task default. The upright weight is zero, so these runs are scored on avoidance and arrival alone.

Runs terminate early once the outcome is decided — reaching the goal, or disqualifying contact. Both conditions are monotone, so early exit does not change any verdict; it makes a 100-trial sweep several times cheaper.

C. Two defects, and their correction

Every run recorded before 2026-08-02 seeded only the initial-state perturbation. The sampling planners’ noise was drawn from system entropy, so two runs of an identical command gave different results. One configuration — avoidance weight 8000, margin 0.08, 50 trials — was measured six times at 6, 8, 10, 13, 14 and 16 solves, a spread of 20 points. Differences smaller than that between single runs of the older sweeps are not findings, and we do not treat them as such.

The sampling planner now takes an explicit noise seed. Every cell of the sweep in Section V-D is seeded and repeated over three seeds, so a table row means *this configuration, this seed* and can be reproduced from the command recorded next to it. The other four planners remain unseeded; their results in Sections V-B and V-C carry the spread described above, which at $n = 100$ is about ± 4 points.

The second defect was in the memoryless control itself, and seeding is what exposed it. Random Sampling is implemented as Predictive Sampling with the incumbent cleared before

each iteration, but the base planner’s first act is to rebuild its plan from the previous iteration’s winner, which discarded the clearing. The control was therefore the planner it was a control for. Two seeded sweeps of the two agreed in all 30 cells, to the trial, which is not a result any pair of distinct planners produces; that is how it was found. Every Random Sampling success rate in this report comes from a run after the correction, and its rows in the lookahead and failure-distribution appendices are dropped rather than restated. Its row in Table IX is the one exception: that measurement predates the fix, and we keep it because the correction adds one plan clear per iteration to a loop whose cost is 99.7% rollouts, leaving what an iteration costs unchanged even though what it computes is not. The wider lesson is procedural: the sweep script recorded the planner it was configured with rather than the one it ran, and printed “predictive sampling” into the manifest of a sweep of a different index. It now reports the name it was given.

D. Intervals and tests

Success counts are binomial in the number of trials, so tables report the standard error $\sqrt{p(1-p)/n}$ and the text reports Wilson score intervals at 95%. Wilson rather than the normal approximation because several cells sit near 0% or above 60%, where the normal interval is poorly calibrated and can extend outside $[0, 1]$. Comparisons between two configurations are two-proportion z -tests. Where several are reported together they are pre-specified, and we say so — the 5×5 grid of Section V-D admits 300 pairwise comparisons, and reporting the largest of them without correction would mean nothing.

One caveat applies to every interval here. Pooling three seeds treats 150 trials as one binomial sample, which understates the spread when the seeds disagree by more than sampling noise: the best cell of Section V-D measures 66, 48 and 48% across its three seeds, a range wider than the pooled interval. The per-seed figures are in the sweep’s `results.csv` and the text gives them wherever the seeds spread by more than about 10 points.

E. Hosts and load

Success rates are insensitive to CPU contention, because the harness counts iterations rather than seconds. Timing is not. Per-iteration timings (Section V-J) were taken with 15 planner threads on an otherwise idle 20-core host. The throughput comparison of Section V-L was taken on an idle Ultra 7 and on an M5 Pro at load average 3.1–4.4; the asymmetry is stated where it bears on a conclusion. The sweep of Section V-D ran five concurrent jobs of four planner threads each on a 20-core host; its wall times and per-iteration figures are contended and are not quoted as speed.

V. RESULTS

A. AgileRRT

The steering primitive ports cleanly. The tree does not close either task.

TABLE III

CORRIDOR, ONE BOTTLENECK. 100 TRIALS, ± 1 S.E. ON THE CLEAN COLUMN. “ ≤ 20 mm” IS THE LAXER CRITERION THIS BENCHMARK USED ORIGINALLY; GRAZING SHARE IS $1 - \text{CLEAN/LAX}$. RANDOM SAMPLING, BELOW THE RULE, IS THE MEMORYLESS CONTROL: 300 SEEDED TRIALS, NO SOLVE UNDER EITHER CRITERION, SO GRAZING SHARE AND MEDIAN SOLVE TIME ARE UNDEFINED.

planner	clean	s.e.	≤ 20 mm	grazing	median t_{solve}
Predictive Sampling	76	4.3	93	18%	2.33 s
PSO (fixed)	51	5.0	66	23%	2.41 s
Annealed Sampling	49	5.0	81	40%	2.53 s
iLQG	35	4.8	46	24%	2.28 s
PSO (stock)	18	3.8	26	31%	2.29 s
Cross-Entropy	2	1.4	21	90%	3.84 s
Random Sampling	0	0.0	0	—	—

On the corridor, a tree of 3000 vertices grown in 379 s never reached the goal set; its best vertex left the cart at $x = 8.03$ m with 5.60 m at the end of the extracted path, against a goal at $x = 6$ that it therefore overshot rather than captured. Of 6008 attempted extends, 50.1% were rejected by collision. On the slalom, 3873 vertices in 1800 s moved the cart to a maximum of 5.83 m of the 11 m required, with 80.5% of 19836 extends rejected by collision.

Where the time goes is the more useful number: 96% of the corridor build and 98% of the slalom build went to nearest-neighbour queries, which are $O(N)$ per extend and therefore $O(N^2)$ over the build. The per-vertex precompute that the paper optimizes was 7% and 2% respectively. This also corrects an earlier extrapolation of ours: a 1000-vertex tree was projected at ≈ 15 s by treating nearest-neighbour as a fixed per-iteration cost, and 1200 vertices measure 125 s.

For the receding-horizon question the conclusion does not depend on the constant. MJPC’s contract is one `OptimizePolicy` call per control step, with a 5 ms budget; a tree build is three orders of magnitude over it. AgileRRT is usable here as an offline trajectory generator, not as an MPC planner, and we do not report it in the planner tables for that reason.

One further observation about the primitive itself. Linearizing about $x_{\text{zero}}(t)$ rather than about x_0 differs only if x_{zero} drifts, and this benchmark starts at the upright equilibrium, which is an equilibrium of the zero-control dynamics: measured drift over 1.0 s is 0.000. At the start state the paper’s contribution is exactly a no-op. From non-equilibrium vertices it is real, but only at horizons beyond $t_h \approx 0.6$ s, where the point linearization saturates the actuator on 100% of steering attempts.

B. One bottleneck

Table III reports 100 trials per planner on the corridor with the objective (7).

Three readings.

The memoryless control is beaten by everything. Random Sampling discards the incumbent every iteration and samples around zero control, and it solves 0 of 300 trials over three seeds (95% CI 0–1.3), colliding in every one. The gap to

TABLE IV

SLALOM, THREE BOTTLENECKS. 100 TRIALS. “GAPS” IS BOTTLENECKS CLEARED BEFORE THE FIRST CONTACT, OUT OF 3; “CORRIDOR” REPEATS THE CLEAN COLUMN OF TABLE III FOR REFERENCE. RANDOM SAMPLING, BELOW THE RULE, IS THE MEMORYLESS CONTROL AT 300 SEEDED TRIALS.

planner	clean	s.e.	≤ 20 mm	gaps	corridor
Predictive Sampling	4	2.0	22	0.94	76
Annealed Sampling	2	1.4	34	0.49	49
PSO (stock)	1	1.0	2	0.26	18
PSO (fixed)	0	0.0	27	0.65	51
iLQG	0	0.0	0	0.09	35
Cross-Entropy	0	0.0	1	0.05	2
Random Sampling	0	0.0	0	0.01	0

Predictive Sampling — which differs from it in nothing but whether the previous iteration’s winner survives into the next — is the largest in the table, larger than the spread across all six other planners combined. Warm starting is not a refinement on this task; at four iterations per decision it is the entire mechanism. Any claim about what a planner’s update rule contributes has to be read against this row, which shows the update rule operating on nothing.

Cross-Entropy fails here rather than underperforming. 2% (95% CI 1–7) with 98% collisions, against 76% for the same sampling machinery under a different update rule. CEM refits a Gaussian to the elite set and samples from it; on a chaotic system the elite set at one iteration does not predict the next, so the refitted covariance concentrates on a region the dynamics have already left. In a rollout the signature is a smooth, committed, wrong trajectory rather than a flailing one.

The collision criterion moves the ranking. Annealed Sampling scores 81 at 20 mm and 49 clean — a 32-point drop, the largest here, and enough to move it from second place to fourth. Grazing share is worth reading on its own: it is the fraction of a planner’s apparent competence that was contact it was not charged for.

C. Three bottlenecks

Table IV is the same protocol on the slalom.

At the default objective nothing solves the slalom. The best planner clears three bottlenecks cleanly in 4 runs out of 100 — 4%, 95% CI 2–10 — and four of the seven never manage it once. The interval on the best row overlaps the interval on every other row: this table reports a wall, not a ranking, and no pairwise test on it would survive a correction for the 21 comparisons it invites.

The partial-credit column does separate the planners, from 0.94 for Predictive Sampling to 0.05 for Cross-Entropy, and that ordering is essentially the corridor ordering. No planner handles chaining better than its single-gap performance predicts. The memoryless control is the exception that proves the point: at 0.01 gaps it is not on the same scale as the others, and the wall this table reports is one only planners that carry a plan get far enough to meet.

The grazing shares here are 82–100%. On the slalom the lax criterion was measuring almost nothing but contact,

TABLE V

SLALOM SOLVE RATE (%) OVER THE AVOIDANCE WEIGHT $\times$ CLEARANCE MARGIN GRID, PREDICTIVE SAMPLING, POOLED OVER THREE PLANNER SEEDS AT 50 TRIALS EACH. ROWS ARE AVOIDANCE WEIGHT, COLUMNS ARE CLEARANCE MARGIN IN METRES. A RULE MARKS A CELL THE SWEEP DID NOT COVER. BOLD IS THE BEST CELL. RANDOM SAMPLING WAS RUN ON THE SAME LOWER-RIGHT CORNER AND IS OMITTED: IT SOLVED ZERO TRIALS IN EVERY CELL (SECTION V-F).

w_{avoid}	0.01	0.02	0.04	0.08	0.12	0.16	0.2
<i>Predictive sampling</i>							
500	—	—	0.7	2.7	3.3	5.3	1.3
2,000	—	—	2.0	5.3	8.0	8.0	2.0
8,000	—	—	6.0	27.3	15.3	9.3	0.7
32,000	—	—	22.7	34.0	10.7	2.0	0.0
128,000	7.3	23.3	54.0	21.3	4.0	0.0	0.0
512,000	28.7	54.0	56.0	—	—	—	—
2.05M	61.3	63.3	44.0	—	—	—	—
8.19M	65.3	52.7	36.0	—	—	—	—

and an earlier version of this work read that column as a ranking inversion between Annealed Sampling and Predictive Sampling. The clean column retracts it.

Why the same first bottleneck gets harder is worth stating, because it is not the geometry: nothing about the gap at $x = 3$ changed. The goal moved from 6 to 11 m, so the Cart residual is roughly $3\times$ larger at the start and the planner commits to more speed on approach. iLQG is the extreme case — 35/100 on the corridor, 0.09 gaps here — arriving at a gap it can otherwise clear with far too much velocity to lay the pendulum out.

D. The weight-margin landscape

Table IV says the default objective does not solve the slalom. It does not say the task is unsolvable by this planner. The avoidance cost has two parameters, and Section III-C argues they are not interchangeable; this section measures them.

We swept $w_{\text{avoid}} \in \{500, 2000, 8000, 32000, 128000\}$ against $\delta \in \{0.04, 0.08, 0.12, 0.16, 0.20\}$ m with predictive sampling, holding the other weights at (7) and using 12 s episodes. Each of the 25 cells was run at three planner seeds $\{1000, 2000, 3000\}$ with 50 trials each, for 150 trials per cell and 3750 runs in total. These are the first five rows of Table V, which reports the mean over seeds; the rows and columns beyond them come from the two later sweeps described in Section V-E and are discussed there.

The surface is not monotone in either parameter, and the two interact. Along the margin axis at fixed weight there is an interior optimum that moves left as the weight rises: 0.16 m at $w = 500$, 0.08 m at $w = 8000$ and 32000, 0.04 m at $w = 128000$. Along the weight axis the benefit of raising the weight depends entirely on the margin — it is worth 53 points at $\delta = 0.04$ and nothing at all at $\delta = 0.20$.

The mechanism is visible in the collision rate. At $w = 128000$, $\delta = 0.20$ the planner collides in 0% of runs and solves 0%: it clears 0.06 gaps on average, meaning it stops before the first bottleneck. That is the wall predicted in Section III-C. With the hinge active across a 0.5 m gap whose half-width is

0.25 m, there is no path through that does not pay, and at a large enough weight the cheapest available behaviour is not to approach. The avoidance term stops being a barrier around each disk and becomes a barrier across the corridor.

The opposite corner fails differently. At $w = 500$, $\delta = 0.04$ the planner collides in 99% of runs. The cost is correct in principle — it charges only inside 4 cm of a disk surface — but at the approach speeds this course induces, 4 cm is a fraction of one 83 ms knot interval of travel. The planner is told about the obstacle after it can no longer steer around it.

The ridge between the two runs from moderate weight with a moderate margin to very high weight with a small one. Its best cell, $w = 128000$ and $\delta = 0.04$, solves 54% (95% CI 46–62; individual seeds 66%, 48%, 48%) against 2.7% (1–7) for the default objective at the same 150-trial budget — $z = 9.87$, $p < 10^{-4}$, which is the one comparison in this grid large enough that no multiplicity correction could touch it. It is also the cell that produced Fig. 1.

E. Past the ridge

Whether that cell is a peak or the near edge of a plateau is not something the grid can answer from its own corner, so we continued past it: weights 512,000, 2.05M and 8.19M against margins 0.01, 0.02 and 0.04 m, again three seeds and 50 trials per seed; these are the lower rows of Table V. The overlapping cell repeats the best cell of the original grid under a later binary and reproduces it seed for seed (66, 48, 48% against 66, 48, 48%), which is the check that the two sweeps can be read as one table.

The optimum keeps moving. The best margin per weight runs $128,000 \rightarrow 0.04$, $512,000 \rightarrow 0.04$, $2.05M \rightarrow 0.02$, $8.19M \rightarrow 0.01$ — the same leftward slide seen in the upper block, continuing without a corner. The best cell of the extension is 8.19M at $\delta = 0.01$ m, at 65% (95% CI 57–72).

Along the ridge itself the gain flattens. Testing the four pre-specified comparisons with two-proportion z -tests:

- the extension’s best against the first grid’s best (128,000/0.04, 54%): $z = 2.00$, $p = 0.045$;
- 128,000/0.04 (54%) against 512,000/0.04 (56%): $z = 0.35$, $p = 0.728$;
- 512,000/0.04 (56%) against 2.05M/0.02 (63%): $z = 1.29$, $p = 0.195$;
- 2.05M/0.02 (63%) against 8.19M/0.01 (65%): $z = 0.36$, $p = 0.718$;

so the $64\times$ span of weight is worth +11 points overall, and 0 of the 3 individual steps along the ridge reaches $p < 0.05$. The last step is +2 points at $p = 0.718$: the ridge is still rising at the edge of the grid, but no longer at a rate this trial budget can resolve. We stopped here rather than continue, because the cost of a further factor of four is another 1800 runs to distinguish two points.

F. The control, on the same corner

The sweep above is predictive sampling throughout, which leaves open whether the ridge is a property of the cost or of that planner. We repeated the extension’s 12 cells with the

TABLE VI
ROLLOUTS PER PLANNING ITERATION, PREDICTIVE SAMPLING AT THE RIDGE CELL $w_{\text{avoid}} = 8.19 \times 10^6$, $\delta = 0.01$ m. ITERATIONS PER CONTROL STEP ARE FIXED AT FOUR FOR EVERY ROW, SO THIS VARIES THE BREADTH OF EACH DECISION, NOT THE NUMBER OF DECISIONS. Δ AND p ARE AGAINST THE 10-ROLLOUT ROW.

rollouts	solved	%	95% CI	per seed	Δ	p
10	98/150	65.3	[57.4, 72.5]	32/35/31	—	—
20	106/150	70.7	[62.9, 77.4]	34/35/37	+5.3	0.322
40	128/150	85.3	[78.8, 90.1]	44/44/40	+20.0	< 0.001
80	126/150	84.0	[77.3, 89.0]	43/40/43	+18.7	< 0.001

incumbent discarded every iteration — same seeds, same trials, same scoring.

It solves nothing: 0 of 1800 trials across all 12 cells (95% CI 0–0.2), and the best any cell manages is 0.02 of three bottlenecks cleared. Because the result is the same zero in every cell, the block is not reproduced in Table V; this paragraph is the whole of it. Sampling afresh around zero control does not reach the first gap, at any weight or margin in this corner.

Two things follow. The tuning that carries predictive sampling from 2.7% to 65% does nothing at all for a planner that discards its plan, so the objective is not solving this task on its own — it is making an already-working search better. And the warm start, not the sampling, is what makes that search work: the only difference between the two planners is whether the previous iteration’s winner survives into the next, and it is worth the entire result.

G. Rollouts per iteration

Every result to this point holds the rollout count at the task file’s 10. The ridge is what the objective can buy; this is what the search width can buy on top of it. All rows sit at the best ridge cell ($w_{\text{avoid}} = 8.19 \times 10^6$, $\delta = 0.01$ m), hold iterations per control step at four, and use three planner seeds of 50 trials.

Forty rollouts solves 85.3% against 65.3% at ten, +20.0 points at $p < 0.001$. Eighty adds nothing: -1.3 points against forty, $p = 0.75$, with the two intervals almost coincident. Twenty is not distinguishable from ten on its own ($p = 0.32$), which is what a minimum detectable effect of 14.3 points at this budget (Section IV-D) predicts for a 5-point difference — the step is real but this experiment could not have resolved it either way.

Two things make the size of this worth stating plainly. It is the second largest effect in the paper, behind only the $2.7\% \rightarrow 65\%$ of the objective tuning and ahead of the entire spread between planners in Table III. And it comes from a single integer in the task file, not from an algorithm. The 10-rollout row also reproduces the corresponding landscape cell seed for seed (32/35/31), so the binary that added the override is neutral.

What saturates at forty is the rollout count, and only at this cell. We did not test whether the plateau moves with the horizon, the knot count, or the objective, so 85% is a ceiling on this axis rather than a ceiling on the task.

TABLE VII

PREDICTIVE AGAINST ANNEALED SAMPLING AT MATCHED ROLLOUT BUDGETS, SAME RIDGE CELL, 150 TRIALS PER ROW. ANNEALED SAMPLING DRAWS ITS TRAJECTORIES ONCE PER ANNEALING ITERATION AND RUNS FOUR, SO ITS BUDGET IS $4\times$ ITS TRAJECTORY COUNT. MS/ITER IS CONTENTED (FIVE CONCURRENT JOBS) AND IS COMPARABLE WITHIN THIS TABLE ONLY.

planner	traj.	rollouts/iter	solved %	collided %	ms/iter
Predictive	10	10	65.3	5.3	1.53
Predictive	40	40	85.3	1.3	6.12
Predictive	80	80	84.0	2.7	10.86
Annealed	10	40	63.3	30.0	6.12
Annealed	40	160	79.3	16.7	19.12

H. Forty rollouts, spent two ways

Forty rollouts per iteration is not a new quantity in this paper. Annealed sampling draws its trajectories once per annealing iteration and runs four of them, so at the task file’s ten trajectories it has been spending forty rollouts per planning iteration throughout — four times what the other samplers spend, in every table up to this point. Table VII lines the two planners up by rollouts instead.

The cost column confirms the $4\times$ independently of any reading of the source: annealed sampling at ten trajectories and predictive sampling at forty report the same 6.12 ms per iteration under the same contention, because both run forty rollouts of the same model.

At a matched forty rollouts, predictive sampling solves 85.3% and annealed sampling 63.3%: 22.0 points, $z = 4.36$, $p < 0.001$. Annealed sampling on forty rollouts is instead indistinguishable from predictive sampling on ten (63.3 against 65.3, $p = 0.72$), which is the sharpest way to state the result — the annealing schedule converts a fourfold rollout budget into nothing measurable.

Giving it four times again does help it: 160 rollouts reaches 79.3%, +16.0 points at $p = 0.002$. It still does not catch predictive sampling at a quarter of that budget (−6.0 points, $p = 0.17$), and it costs 19.1 ms per iteration against 6.1, roughly four times the control period. There is no thread count that fixes this; MJPC parallelizes over rollouts within an annealing iteration, and the four annealing iterations are sequential by construction, since each re-centres its noise on the previous one’s result.

The collision column identifies the failure. At matched budget annealed sampling collides on 30.0% of trials against 1.3%, and at 160 rollouts it is still at 16.7%. It is not running out of samples or of time — the harness hands out iterations regardless of cost — it is committing to trajectories that hit disks, and extra samples reduce that only slowly.

The reduction step is the candidate explanation. Predictive sampling keeps the single best rollout; annealed sampling keeps an MPPI-weighted average over all of them. Around a disk the feasible set is disjoint — pass left or pass right — and a weighted average of a left sample and a right sample is a plan through the middle. That predicts exactly what the column shows: a planner that collides much more often than

TABLE VIII

SLIDING PLAN OFF AGAINST ON, AT THE THREE BEST CELLS OF THE RIDGE, 150 TRIALS PER ARM PER CELL. “COL” IS THE COLLISION RATE. OFF IS THE DEFAULT AND IS WHAT EVERY OTHER TABLE IN THIS PAPER USES.

w_{avoid}	δ	off		on		Δ	p
		%	col	%	col		
8.19×10^6	0.01	65.3	5.3	69.3	14.0	+4.0	0.460
2.05×10^6	0.02	63.3	4.0	70.7	6.7	+7.3	0.177
2.05×10^6	0.01	61.3	25.3	45.3	48.0	−16.0	0.005
pooled		63.3		61.8		−1.6	0.630

the one it outsamples, and that improves with more samples without ever fixing the underlying step, since more samples make the average better estimated rather than less averaged. We did not test this directly; the distinguishing experiment is annealed sampling with the MPPI update replaced by an argmax, holding the annealing schedule fixed.

This revises how Table III should be read. Annealed sampling’s 49/100 there was not a fair-budget loss but a $4\times$ -budget loss, and correcting for the budget widens the gap rather than closing it.

I. Carrying the plan forward

Section V-E establishes that discarding the incumbent costs everything. The sliding plan is the other end of the same question: instead of resampling the nominal from the previous winner, shift the existing plan forward by one control step and extend it. It is a flag in the sampling planner, off by default, and it changes what the search is handed each iteration without changing the search.

Pooled over the three cells it is a wash (−1.6 points, $p = 0.63$). The cells disagree in a way the pooled number hides: two lean positive without reaching significance, and the third loses 16.0 points at $p = 0.005$. The collision column gives the mechanism. Sliding raises the collision rate in all three cells ($5.3 \rightarrow 14.0$, $4.0 \rightarrow 6.7$, $25.3 \rightarrow 48.0$), and at the cell where the baseline was already collision-prone that increase is large enough to swamp whatever the carried plan was buying. A shifted plan was optimal for the obstacle geometry one step ago; resampling from the winner re-fits it to the geometry now. Near a disk, one control step of staleness is expensive.

So the warm start matters in the sense of Section V-E — the planner needs a plan to start from — but not in the sense that a more faithfully carried plan is a better one. All three off arms reproduce their landscape cells seed for seed.

J. What an iteration costs

The success tables hold iterations constant. Table IX is the other half: what an iteration costs, and therefore whether a planner could deliver those iterations outside the harness. Measurements are 6000 planning iterations per planner with early exit disabled, 15 planner threads on an idle 20-core host.

The four 10-rollout samplers are within 3% of each other on the mean, and almost all of that spread is contact: with

TABLE IX

PER-ITERATION WALL TIME, CORRIDOR. “CONTACT-FREE” IS THE SAME MEASUREMENT ON THE BALANCE STAGE, WITH OBSTACLES REMOVED. THE BUDGET IS 5 MS.

planner	ms/iter	p95	worst	% budget	contact-free
Random Sampling	1.000	1.59	4.6	20%	0.924
PSO	1.018	1.64	7.8	20%	0.926
Predictive Sampling	1.031	1.62	6.4	21%	0.939
Cross-Entropy	1.326	2.27	6.3	27%	0.986
Annealed Sampling	4.128	5.91	14.5	83%	3.779
iLQG	4.651	7.38	17.9	93%	4.343

obstacles removed they converge to 0.92–0.99 ms. Cross-Entropy’s apparent 30% overhead is not the algorithm but its 79% collision rate — MuJoCo charges more for contact-rich rollouts, so a planner that drives into disks pays twice, once in the score and once on the clock. Per-iteration cost is essentially identical across the sampling family; only the outcome differs.

The two exceptions are structural. Annealed Sampling’s $4.0\times$ is its annealing multiplier, measured independently of the code: 99.9% of its iteration is rollouts, so there is no overhead to trim — the cost is the extra samples. iLQG spends only 25% of its iteration on rollouts; 45% goes to finite-difference model and cost derivatives and 27% to the nominal rollout, while the Riccati backward pass, the part usually expected to dominate, is 2.7%. On a 4-DOF system the derivatives are already the bill, which is why iLQG’s cost will scale with model size worse than any sampler’s here.

Every planner fits the 5 ms budget on the mean. On the tail, iLQG (p95 7.4 ms) and Annealed Sampling (p95 5.9 ms) do not: a real loop would drop iterations for them precisely when the state is hardest, so their success rates in Tables III and IV are optimistic in a way the 1 ms planners’ are not.

K. Control effort and cost share

Table X characterizes what the configurations actually do with the actuator, averaged over eight recorded rollouts each. It answers a question the success rate cannot: whether two configurations with similar scores are doing the same thing. The lower block walks the ridge of Section V-D in ascending solve rate, so the table can be read as a progression rather than a list of near-misses.

Three patterns. First, actuator saturation is negligible everywhere (0–1% of steps). The 20 N limit is not the binding constraint on this task at any configuration we measured — the binding constraint is what the planner can find, not what the actuator can deliver.

Second, the avoidance share *falls* along the ridge, from 63% at its start to 29% at its best cell, while the solve rate rises from 54% to 65%. The natural expectation is the opposite: a $64\times$ larger avoidance weight should dominate the running cost more, not less. It does not, because the margin shrinks as the weight grows — the ridge slides from $\delta = 0.04$ to $\delta = 0.01$ — and the hinge (5) is zero outside the margin band. A stiff barrier over a 1 cm band is paid on far fewer steps than a soft one over 4 cm. The best configuration measured here is the

TABLE X

CONTROL CHARACTER AND COST SHARE, MEANS OVER 8 ROLLOUTS PER CONFIGURATION. “SAT” IS THE PERCENTAGE OF STEPS AT THE 20 N LIMIT; EFFORT IS $\int u^2 dt$; REVERSALS IS SIGN CHANGES OF u PER SECOND. COST SHARES ARE OF THE MEAN TOTAL RUNNING COST. THE LOWER BLOCK IS THE RIDGE, WITH ITS SLALOM SOLVE RATE FROM TABLE V.

configuration	solved	sat	$\overline{ u }$	effort	rev/s	Cart	Avoid
PS $w500 \delta.08$	2.7	0	4.55	535	6.0	38%	47%
PS $w8k \delta.08$	27.3	1	5.97	830	7.8	18%	75%
PS $w8k \delta.20$	0.7	1	9.07	1395	14.8	47%	46%
PS $w32k \delta.20$	0.0	1	9.02	1380	15.3	39%	55%
PSO $w8k \delta.20$	—	1	6.97	926	13.2	48%	46%
AS $h1 w500$	—	0	3.17	322	10.9	43%	45%
AS $h2 w500$	—	0	2.74	248	16.3	40%	48%
AS $h2 w8k \delta.20$	—	0	4.10	334	6.5	41%	54%
PS $w128k \delta.04$	54.0	0	4.56	577	6.7	28%	63%
PS $w512k \delta.02$	54.0	1	7.24	1053	11.3	30%	59%
PS $w2.05M \delta.02$	63.3	0	6.60	930	10.7	33%	57%
PS $w8.19M \delta.01$	65.3	1	7.07	1027	11.3	56%	29%
RS $w8.19M \delta.01$	0.0	0	4.18	452	98.3	83%	0%

one that pays the least avoidance cost, because it is the one that stays out of the band; the weight buys a sharper boundary, not a larger toll.

Third, the memoryless control is a distinct failure mode rather than a weaker version of the same behaviour. At 98.3 reversals per second it changes the sign of the actuator on roughly every other control step, spends 83% of its cost on the cart term and 0% on avoidance — it never gets near a disk — and uses less than half the effort of the planner it is a control for. It is not steering badly; it is not steering.

The annealed-sampling rows are the low-effort outliers, at 248–334 against 322–1395 elsewhere. Its MPPI-style weighted average over 40 rollouts is a smoother control signal than a winner-take-all pick over 10, which is visible here and does not translate into a better score.

L. Planning throughput across hosts

Table XI reports one fixed workload — predictive sampling, 1.0 s horizon, 12 knots, 8000 iterations, early exit disabled — at two rollout counts and several thread counts. The two counts are the ones Table VI compares, so this section says what the 20-point gain costs.

Per core the M5 Pro is the faster machine, by 29–37% depending on which budget the comparison is drawn from (4.162 against 3.027 ms at 10 rollouts, 16.359 against 12.673 at 40). At 10 rollouts that advantage does not survive contact with the thread pool: both hosts flatten at 1.21–1.25 ms, within 3% of each other, having converted $3.3\times$ and $2.5\times$ of their cores respectively. This is the result the earlier version of this table reported, and it is unchanged — ten rollouts of 201 steps is not enough parallel work to distinguish two machines of this class.

At 40 rollouts they separate, and in the direction the single-thread numbers do not predict. The Ultra 7 improves, from $3.3\times$ to $4.9\times$: MJPC parallelizes over rollouts, so 10 rollouts

TABLE XI

PER-ITERATION WALL TIME IN MS BY PLANNER THREAD COUNT, AT TWO ROLLOUT BUDGETS ON TWO HOSTS; PARENTHESED FIGURES ARE SPEEDUP OVER THE SAME COLUMN’S ONE-THREAD ROW. ULTRA 7 IDLE, M5 PRO AT LOAD AVERAGE 3.1 AND 4.4. BOLD IS EACH COLUMN’S BEST. SINGLE-THREAD `mj_step` THROUGHPUT IS A PROPERTY OF THE CORE AND MUST AGREE ACROSS THE TWO BUDGETS OF ONE HOST; IT DOES, AT 0.48 AND 0.49 MSTEPS/S ON THE ULTRA 7 AND 0.66 AND 0.63 ON THE M5 PRO.

thr.	Intel Core Ultra 7 265KF		Apple M5 Pro	
	10 rollouts		40 rollouts	
	ms	ms	ms	ms
1	4.162	16.359	3.027	12.673
2	2.207 (1.9)	8.741 (1.9)	1.802 (1.7)	7.074 (1.8)
4	1.562 (2.7)	4.957 (3.3)	1.299 (2.3)	6.138 (2.1)
8	1.319 (3.2)	3.399 (4.8)	1.264 (2.4)	7.257 (1.7)
12	1.251 (3.3)	3.317 (4.9)	1.213 (2.5)	7.130 (1.8)
16	1.395 (3.0)	3.400 (4.8)	1.271 (2.4)	6.595 (1.9)

cannot occupy more than 10 threads however many the host has, and 40 gives the pool enough work to be worth filling. The M5 Pro moves the other way, from $2.5\times$ to $2.1\times$, and its curve stops being a plateau — 4 threads is its best point, 8 and 12 are *worse* than 4, and 16 recovers part of the loss. The Ultra 7 ends 46% faster in absolute terms (3.317 against 6.138 ms) on the machine that is a third quicker per core.

A plausible mechanism is core heterogeneity. MJPC joins its rollout threads every iteration, so an iteration costs what the slowest thread costs, and on a performance/efficiency split the threads landing on efficiency cores are stragglers. The penalty should scale with rollouts per thread, which is what the two budgets show: at 10 rollouts and 8 threads a straggler is one rollout behind, at 40 rollouts it is five. We have not confirmed this — it predicts that the best thread count equals the performance-core count, and that pinning to those cores removes the inversion, neither of which we tested. The M5 Pro rows were also taken at load average 3.1 and 4.4 rather than idle, so the non-monotonicity in particular should be treated as provisional.

The consequence for Table VI is that the price of the 20-point gain is host-dependent. On the Ultra 7 the $4\times$ rollout increase costs $2.65\times$ wall time rather than $4\times$ ($1.251 \rightarrow 3.317$ ms), part of it paid by parallelism that 10 rollouts was leaving on the floor, and at 3.32 ms mean and 3.88 ms p95 it fits the 5 ms budget — on the tail as well as the mean, unlike annealed sampling, whose p95 of 5.91 ms exceeds the period (Table IX). That matters because the tail iterations are the ones near a gap. On the M5 Pro the same increase costs $5.1\times$ ($1.213 \rightarrow 6.138$ ms) and does not fit the budget at any thread count we measured, mean or p95 (7.39 ms at its best point). The configuration that solves 85% is runnable on one of these two hosts and not the other, which is not a distinction either machine’s single-thread throughput would have predicted.

For calibration on contention: the same 10-rollout workload on the Ultra 7 at load average 9.96 reports 4.145 ms single-threaded and 1.182 ms at 10 threads, within 0.3% of the quiet measurement. On this host the throughput numbers are robust

to moderate background load. Success rates never depended on it.

VI. DISCUSSION

A. Why *iLQG* does poorly

iLQG is the weakest planner on the corridor other than stock PSO and CEM, and it is last on the slalom’s partial-credit column at 0.09 gaps. Three contributions, in order of size.

The linearization is invalidated by exactly the event that matters. *iLQG* builds a first-order model of f around a nominal trajectory. Contact makes f discontinuous, and the obstacles here are real collision geoms, so the derivative that the backward pass uses is a model of a world in which the disks do not exist until the nominal trajectory touches one — at which point the finite-difference Jacobian across a contact event is not a useful local model either. Sampling planners are indifferent to this: a rollout that hits a disk returns a large number, which is all they need.

The cost structure penalizes it twice. Its derivatives are 45% of a 4.65 ms iteration, so it is $4.5\times$ the cost of predictive sampling for the same iteration count; on the tail it exceeds the 5 ms budget, meaning the harness credits it with iterations a real loop would not deliver.

And its failure mode on the slalom is specific: it arrives at the first gap far too fast. With the goal at 11 m, the Cart residual dominates early, and a first-order method following that gradient accelerates; a sampler with 8 N of exploration noise stumbles into slower approaches that a descent direction does not offer. This is a case where the smoothness assumption is not merely unhelpful, it selects the wrong behaviour.

B. Why *AgileRRT* does not fit

The negative result in Section V-A is not about the steering primitive, which ports to MuJoCo cleanly and cheaply: `mjd_transitionFD` replaces the analytic Jacobians, `mj_step` replaces the ODE integrator, and MuJoCo’s contacts replace the hand-rolled swept line-circle tests. It is about the search. A tree build is an offline computation of a single open-loop path; MJPC’s contract is a bounded computation per control step that returns a policy for the current state. Three orders of magnitude separate them, and 96% of the tree build is spent in nearest-neighbour queries that grow as $O(N^2)$, so the gap widens with tree size rather than closing.

There is a real observation underneath the comparison. *AgileRRT*’s extends were rejected by collision 50% of the time on the corridor and 80% on the slalom. The obstacles are hard for a sampling-based planner in configuration space for the same reason they are hard for a sampling-based MPC: the feasible set is thin relative to the reachable set, and neither method has a mechanism that concentrates samples in it. The MPC’s avoidance cost is such a mechanism, which is why Section V-D moves the success rate by an order of magnitude and no amount of tree growth does.

C. Getting to real-time reliability

The best measured configuration on the slalom solves 85%, at 3.32 ms per iteration against a 5 ms budget on one of our two hosts and 6.14 ms on the other — runnable, but not everywhere, and still failing about one run in seven; the planners at the default objective solve a few percent. Four directions, ordered by how much the measurements above support them.

Tune the cost before the optimizer. The largest single effect we measured is Section V-D's $2.7\% \rightarrow 65\%$, from changing two numbers in the objective. That is a bigger effect than the entire spread between planners in Table III, and it is the one result here with a p -value that no correction could touch. It also appears to be nearly spent: the last $4\times$ of weight bought 2 points at $p = 0.72$, so a further factor of four is not where the next 35 points are.

Shape the residual rather than reweighting it. The hinge (5) is symmetric in the direction of approach — it charges the same for a head moving toward a disk as away from it, and the same for a head near the gap centreline as one about to clip an edge. A residual that charges on predicted time-to-contact rather than on distance would keep the cost at zero for a head leaving the gap, which is what makes the wide-margin configurations fail. This is untested.

Spend the budget where the run is decided. Fig. 1 makes the case: the outcome is decided in the roughly 80 ms around each crossing, and the planner spends the same four iterations per control step for the other 11.7 s. Nothing in MJPC's interface prevents a task from reporting a scalar urgency and the harness from varying the iteration count with it. At 1.03 ms per iteration and a 20 ms control period at $s = 0.25$, there is roughly $4\times$ headroom to redirect.

Raise the rollout count, not the planner sophistication. This one we ran, and it is the largest remaining effect: $65.3 \rightarrow 85.3\%$ for $10 \rightarrow 40$ rollouts (Table VI), at $2.65\times$ the wall time rather than $4\times$, because 10 rollouts could not fill the thread pool in the first place (Table XI). It fits the 5 ms budget at 12 threads with a third of the period spare on the Ultra 7 and does not fit it at all on the M5 Pro, so this direction buys reliability at the cost of a portability constraint the 10-rollout configuration did not have. It is also the same $4\times$ that annealed sampling has been spending all along, and Table VII shows the two ways of spending it are not close: at matched rollouts the plain sampler is 22 points ahead, and the annealed one needs 160 rollouts and 19 ms an iteration to get within six points of what the plain one does on forty. Eighty rollouts buys predictive sampling nothing further, so this axis is now spent as well.

Carry the plan better. The memoryless control solves nothing — 0 of 300 on the corridor, 0 of 1800 across the ridge corner — against 76% and 65% for the same sampler with the incumbent kept. Whatever is producing these results lives in the warm start, and no experiment here measures it by degree: the two planners differ by all of it or none of it. Shrinking the incumbent toward zero by a factor ρ

before each iteration would interpolate between them, and the shape of that curve distinguishes two mechanisms with different fixes — a plan that merely seeds the search, or one that accumulates refinement across iterations. What the interpolation would *not* settle is whether a more faithfully carried plan is better, because Table VIII already answers that: shifting and extending the existing plan rather than resampling it is a wash pooled and a 16-point loss where collisions were already common. The incumbent is worth having and is not worth preserving exactly.

What we would not pursue is a more elaborate zeroth-order update rule. Table III is the argument, though not the one we expected to make from it: Cross-Entropy at 2% and stock PSO at 18% sit far below plain predictive sampling at 76%, so on this task added machinery has consistently cost rather than paid. Table VII closes the one loophole in that argument. Annealed sampling was the elaborate rule with the best case for itself, since every previous table understated its budget rather than overstating it; measured at matched rollouts it loses by 22 points, and the correction moves it further behind rather than closer. The margin to recover is in what the search is handed each iteration, not in how it folds the winner back in.

VII. CONCLUSION

On a chaotic, underactuated obstacle-avoidance task, six sampling and first-order MPC planners at equal iteration budget span 2 to 76 solves out of 100 on a single bottleneck and 0 to 4 out of 100 on three chained bottlenecks. The ordering is stable across both. A memoryless control — the same sampler with its incumbent discarded each iteration — solves none of 300 trials on the single bottleneck and none of 1800 across the best corner of the parameter grid, which puts the warm start, not the update rule, at the centre of what these planners do. Retuning two parameters of the avoidance cost — its weight and the clearance margin at which it activates — raises the three-bottleneck rate from 2.7% to 65%, a larger effect than the entire spread between planners, and the sweeps that establish it show the two parameters interacting rather than acting independently: the optimum is a ridge whose best margin falls as the weight rises, and tuning either knob alone crosses it instead of following it. Widening the search from 10 rollouts per iteration to 40 adds a further 20 points, to 85%; 80 adds nothing. That increase costs $2.65\times$ the wall time rather than $4\times$ on a 20-core x86 host, because at 10 rollouts the planner could not fill the thread pool, and it still fits the budget there. It costs $5.1\times$ on a 15-core Apple Silicon host and does not, which is a distinction neither machine's single-core throughput predicts and which only appears once the rollout count is large enough to expose how each host schedules a parallel join. Per-iteration cost is within 3% across the sampling family and the 5 ms budget is met on the mean by all six planners, so the differences reported here are differences of algorithm and not of throughput. The two largest effects we found were an objective parameter and a rollout count, and the smallest were the choices of update rule; on this task the search was never the bottleneck. It remains unsolved in the

sense that matters for deployment: the best configuration still fails about one run in seven, and both knobs that produced the gains are now at the point where a further factor of four buys nothing measurable.

REPRODUCING

Every number in this report comes from a seeded run whose command line, host, commit and binary checksum are recorded in a manifest beside its log. The benchmark harness, the sweep scripts, and the task definition are in the repository under `mjpc/tasks/triple_pendulum_cartpole/`.

Table XI is the one measurement that is a property of the host rather than of the algorithm, so it is the one worth repeating elsewhere. It is produced by `benchmark/machine_bench.sh`, which takes an output directory and a thread list, and reads the rollout count from TRAJ:

```
TRAJ=10 machine_bench.sh out/m10 1 2 4 8 12
16
TRAJ=40 machine_bench.sh out/m40 1 2 4 8 12
16
```

Everything else is fixed inside the script — predictive sampling, 1.0 s horizon, 12 knots, `total_time=2.0`, 5 repeats, `early_exit=false`, `seed=1` — so two hosts run the identical workload rather than a similar one. Run it with nothing else on the machine. Compare the one-thread `ms/iter` across hosts for per-core speed, and the `Msteps/s` column relative to its own one-thread value for how much of the extra cores each host converts. Single-thread `Msteps/s` must agree between the two invocations; if it does not, the rollout override did not reach the planner.

REFERENCES

- [1] T. Howell, N. Gileadi, S. Tunyasuvunakool, K. Zakka, T. Erez, and Y. Tassa, “Predictive sampling: Real-time behaviour synthesis with MuJoCo,” arXiv:2212.00541, 2022.
- [2] E. Todorov, T. Erez, and Y. Tassa, “MuJoCo: A physics engine for model-based control,” in *IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2012, pp. 5026–5033.
- [3] T. M. Caldwell and N. Correll, “Fast sample-based planning for dynamic systems by zero-control linearization-based steering,” in *Int. Symp. Robotics Research (ISRR)*, 2015.
- [4] R. Y. Rubinstein and D. P. Kroese, *The Cross-Entropy Method*. New York: Springer, 2004.
- [5] J. Kennedy and R. Eberhart, “Particle swarm optimization,” in *Proc. IEEE Int. Conf. Neural Networks*, 1995, pp. 1942–1948.
- [6] H. Xue, C. Pan, Z. Yi, G. Qu, and G. Shi, “Full-order sampling-based MPC for torque-level locomotion control via diffusion-style annealing,” arXiv:2409.15610, 2024.
- [7] G. Williams, A. Aldrich, and E. A. Theodorou, “Model predictive path integral control: From theory to parallel computation,” *J. Guidance, Control, and Dynamics*, vol. 40, no. 2, pp. 344–357, 2017.
- [8] Y. Tassa, T. Erez, and E. Todorov, “Synthesis and stabilization of complex behaviors through online trajectory optimization,” in *IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2012, pp. 4906–4913.
- [9] S. M. LaValle and J. J. Kuffner, “Randomized kinodynamic planning,” *Int. J. Robotics Research*, vol. 20, no. 5, pp. 378–400, 2001.

APPENDIX A LOOKAHEAD

The bottlenecks are 3 m apart, so at the 1 s horizon a planner can barely see the next gap while committing to the posture for this one. Lengthening the horizon makes things worse, and

the controlled version says why. Success rate is useless as the measure (every cell is 0–6 of 50), so the figure is partial credit, 50 trials each.

TABLE XII
GAPS CLEARED BEFORE FIRST CONTACT, OUT OF 3.

horizon	knots	spacing	Pred. Sampling
1.0 s	12	83 ms	0.96
2.0 s	12	167 ms	0.48
3.0 s	12	250 ms	0.26
1.0 s	12	83 ms	0.98
2.0 s	24	83 ms	0.80
3.0 s	36	83 ms	0.86

With the knot count fixed at 12, tripling the horizon triples the knot spacing, so the planner buys lookahead by coarsening the action it is about to execute. Scaling knots with the horizon (lower block) removes the collapse entirely — and does not improve on the 1 s row either. On this task the binding constraint is not what the planner can see; it is that a chaotic three-link pendulum cannot be steered accurately enough over 3 m to arrive at the next gap laid out, however far ahead the plan extends. The practical consequence is that `agent_horizon` and `sampling_spline_points` are not independent knobs and sweeping one without the other measures their product.

APPENDIX B WHY NOT COARSEN THE TIMESTEP

A 20 ms physics timestep produces the same 50 Hz wall-clock control rate as `--speed=0.25` and looks like the simpler way to get there. It is a different experiment and loses on every axis. It does not buy planner compute: $\Delta t = 5$ ms at $s = 0.25$ gives four iterations per 20 ms decision, $\Delta t = 20$ ms at $s = 1.0$ gives one. It does not buy wall time either, since the sweeps are planner-bound and physics is already under 1% of the run. It corrupts the dynamics measurably: with $u = 0$ and no friction, horizontal momentum is conserved, so centre-of-mass drift is pure integrator error.

TABLE XIII
CENTRE-OF-MASS DRIFT OVER 12 S OF THE ZERO-CONTROL SWING, WHICH IS AN EXACT INVARIANT OF THE TRUE DYNAMICS.

integrator	Δt	CoM _x drift
implicitfast	0.005 (the task)	0.088 m
implicitfast	0.010	0.326 m
implicitfast	0.020	1.476 m
rk4	0.005	0.0002 m

At $\Delta t = 0.020$ that is 1.48 m of cart travel manufactured from nothing, on a course whose half-gap is 0.25 m. And it silently loosens the collision test, which is sampled once per `mj_step`: across 256 recorded slalom rollouts, 46% of the contact episodes that first disqualify a run last ≤ 4 steps (20 ms), so a 20 ms timestep steps over a large share of them and reports a higher success rate for a worse simulation. The

RK4 row is included only to show how much of the drift at $\Delta t = 0.005$ is the integrator rather than the timestep; it is not a recommendation, because RK4 re-evaluates the dynamics at sub-steps, which is where MuJoCo’s contact impulses are discontinuous.

APPENDIX C

FAILURE DISTRIBUTION

Bottlenecks cleared before first contact, 40 trials per planner, recorded with early exit disabled so each run continues past its first contact. Counting the whole rollout instead would mark every trial as reaching all three, since the cart keeps driving after contact and parks at the rail limit.

TABLE XIV
DISTRIBUTION OF GAPS CLEARED BEFORE FIRST CONTACT.

planner	0	1	2	3	clean solves
Predictive Sampling	11	23	3	3	3
PSO	17	22	1	0	0
Annealed Sampling	28	12	0	0	0
Cross-Entropy	38	2	0	0	0
iLQG	39	1	0	0	0

The distributions are unimodal at 0 or 1 gap for every planner. Nothing here clears two gaps reliably and then fails at the third; the failure is at the first or second crossing, which is consistent with the reading in Section V-C that chaining is not a separate capability from threading one gap. Random Sampling is omitted from this table and from Table XII: its rows in both predate the correction described in Section IV-C, and its distribution is in any case degenerate — it clears 0.00–0.02 gaps on average over 1800 trials, so every trial falls in the first column.